

Ranhui Yan

Teaching Assistant and Machine Learning Researcher

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Research Profile

I work on applied AI, using graph learning and deep learning to study interdisciplinary problems. My research focuses on graph neural networks, particularly spectral methods, heterogeneous graphs, and robustness under structural perturbations.

Academic Interests

- Applied AI: graph and deep learning for interdisciplinary problems
- Graph neural networks, heterogeneous graphs, and spectral graph learning
- Robustness in graph learning

Experience

Teaching Assistant

2024–Present

School of Computer and Artificial Intelligence, Guangzhou Xinhua University, Guangzhou, China

- Teach and support courses in Calculus, Linear Algebra, Statistics, and Machine Learning.

Education

M.S. in Statistics

2021–2024

Guangdong University of Finance & Economics, Guangzhou, China
Supervised by Prof. Jia Cai.

B.S. in Information and Computing Science

2017–2021

China Jiliang University, Hangzhou, China

Selected Publications

R. Yan, J. Cai, M. Chen, and H. Yang. "Convex–Concave Quadratic Spectral Filtering for Graph Neural Networks." *Accepted at ECML 2026*. [arXiv:2606.24956](#). [Code](#).

R. Yan and J. Cai. "Virtual Nodes based Heterogeneous Graph Convolutional Neural Network for Efficient Long-Range Information Aggregation." *International Conference on Artificial Neural Networks (ICANN)*, 2024, pp. 217–231. [arXiv:2509.23660](#). [DOI](#). [Code](#).

Z. Xiong, J. Cai, **R. Yan**, and X. Huang. "Noise Perturbation Based Graph Contrastive Learning via Flexible Filters for Node Classification." *International Joint Conference on Neural Networks (IJCNN)*, 2024, pp. 1–10. [DOI](#).

R. Yan and J. Cai. "An End-to-End Dense Connected Heterogeneous Graph Convolutional Neural Network." *International Conference on Neural Information Processing*, 2023, pp. 462–475. [DOI](#).